



**Department of Electrical & Electronics
Engineering**

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**V Semester
RTOS Course Project**

Report on

**OT AIR GUARD: Real-Time
Air Quality Monitoring System for Sterile
Environments**

Project Associates

Name	USN
Ankush S Jalasandi	01FE23BEE035
Shivkumar Bannad	01FE23BEE037
Samarth Shirahatti	01FE23BEE039
Animesh	01FE23BEE009

Abstract

Maintaining sterile air quality in hospital Operating Theaters (OTs) is critical for preventing Surgical Site Infections (SSIs) and ensuring patient safety. According to standards like ASHRAE 170, OTs require strict control over ventilation, pressure, and contaminant levels. Conventional monitoring systems often rely on manual checks or simple polling-based microcontrollers, which can suffer from response delays and inability to handle multiple critical tasks simultaneously. This project, "OT AIR GUARD," proposes a robust, Real-Time Operating System (RTOS) based solution designed to continuously monitor and control the air quality parameters within an OT environment using the LPC1768 Cortex-M3 microcontroller.

The system integrates two primary analog sensors: the MQ-2 sensor for detecting smoke, combustible gases, and anesthesia leaks, and the MQ-135 sensor for monitoring Air Quality Index (AQI) and Volatile Organic Compounds (VOCs) such as sanitizers and perfumes. To ensure deterministic and reliable performance, the firmware is built upon the RTX Kernel. The application is partitioned into three independent tasks: a Data Acquisition Task (Producer), a Control Logic Task (Consumer), and a Display Interface Task.

A key innovation in this project is the use of **Semaphore Synchronization** to optimize CPU usage. Instead of the Control Task checking for data using inefficient time delays, it remains in a "blocked" state until the Data Acquisition Task signals that fresh sensor data is ready. This event-driven architecture ensures that the system reacts instantly to hazardous conditions without wasting processor cycles. When contamination levels exceed safety thresholds (e.g., MQ-2 > 70% or MQ-135 > 60%), the system automatically triggers a 2-channel relay module. Channel 1 activates an Exhaust Fan to flush out contaminants, while Channel 2 activates a Piezo Buzzer to alert medical staff immediately. The status of the environment (SAFE, NORMAL, DANGER) is displayed in real-time on a 16x2 LCD. The result is a highly responsive, automated safety guard that minimizes human error and enhances the biological safety of the surgical environment.

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Chapter 1: Introduction

1.1 General Introduction

Air quality in healthcare facilities, particularly in Operating Theaters (OTs), is a matter of life and death. Airborne contaminants, including smoke from electrocautery devices, leaking anesthesia gases, or excessive sterilization chemicals (VOCs), pose significant health risks to both patients and surgical teams. Standard HVAC systems provide general ventilation but often lack the localized, rapid-response capability to detect sudden spikes in hazardous gases. This project aims to bridge that gap by developing a dedicated "OT Air Guard" that uses embedded intelligence to monitor and react to air quality threats in real-time.

1.2 Existing Methods

Currently, many hospitals rely on:

1. **Manual Sensing:** Periodical checks by facility managers, which are prone to human error and data gaps.
2. **General HVAC Building Management Systems (BMS):** These monitor broad temperature and humidity but often lack specific gas detection capabilities within the surgical zone.
3. **Super-Loop Architectures:** Simple microcontroller projects that run in a single while(1) loop. These systems suffer from "blocking delays"—if the processor is busy updating a slow LCD screen, it might miss a critical gas spike or delay the alarm trigger.

Chapter 2: Literature Survey & Objectives

2.1 Literature Survey

- **Study 1:** Research on "IoT based Indoor Air Quality Monitoring" highlights the effectiveness of MQ-series sensors for detecting VOCs but notes the latency issues in cloud-based reporting.
- **Study 2:** "Real-Time Systems in Healthcare" emphasizes the need for deterministic responses (hard deadlines) in medical devices, suggesting RTOS as a superior alternative to bare-metal programming.
- **Study 3:** ASHRAE Standard 170 guidelines specify the need for positive pressure and high air change rates (ACH) in OTs, providing the theoretical basis for our exhaust fan control logic.

2.2 Problem Statement

Hospitals depend heavily on medical gases like oxygen and nitrous oxide, but even small leaks in confined areas such as operating theaters can cause serious hazards, including fire, oxygen enrichment, or toxic exposure. Existing monitoring systems are often costly and rely on sequential processing, which can introduce delays and fail to detect simultaneous events in real time. This project addresses the need for a cost-effective, real-time medical gas monitoring system. By using an RTOS-based design with preemptive multitasking, the system ensures immediate detection and response to gas leaks, prioritizing safety-critical actions such as alarms and exhaust activation over non-essential tasks, thereby enhancing hospital safety and reliability.

2.3 Objectives

1. To interface MQ-2 and MQ-135 analog sensors with the LPC1768 ADC.
2. To implement an RTOS-based firmware using the RTX Kernel with multiple tasks.
3. To utilize **Binary Semaphores** for efficient inter-task communication and synchronization.
4. To control high-voltage actuators (Fan/Alarm) via a Relay module based on calibrated danger thresholds.

2.4 Task Scheduling Gantt Chart

2.4.1 Scheduling Policy

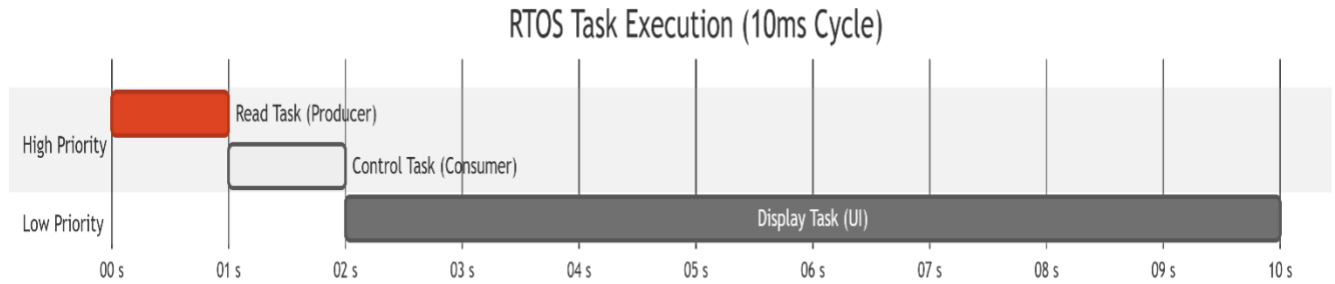
The firmware utilizes the keil-RTX Real-Time Operating System, which employs a Preemptive Priority-Based Scheduling algorithm. In this project, three tasks are defined with specific priorities to ensure deterministic behavior:

- Task 1 (Read_Task): Priority 3 (High) - Critical Data Acquisition.
- Task 2 (Control_Task): Priority 3 (High) - Safety Critical Logic.
- Task 3 (Display_Task): Priority 2 (Normal) - Non-critical User Interface.

The system uses Binary Semaphore Synchronization (sem_data_ready) to coordinate Task 1 and Task 2. This prevents CPU wastage by keeping the Control Task in a BLOCKED state until fresh data is available.

2.4.2 Execution Timeline (Gantt Chart)

The following timing diagram illustrates one complete cycle of operation (approx. 10ms window). It demonstrates how the Semaphore signal triggers an immediate Context Switch from the Acquisition task to the Control task, ensuring zero latency for safety checks.

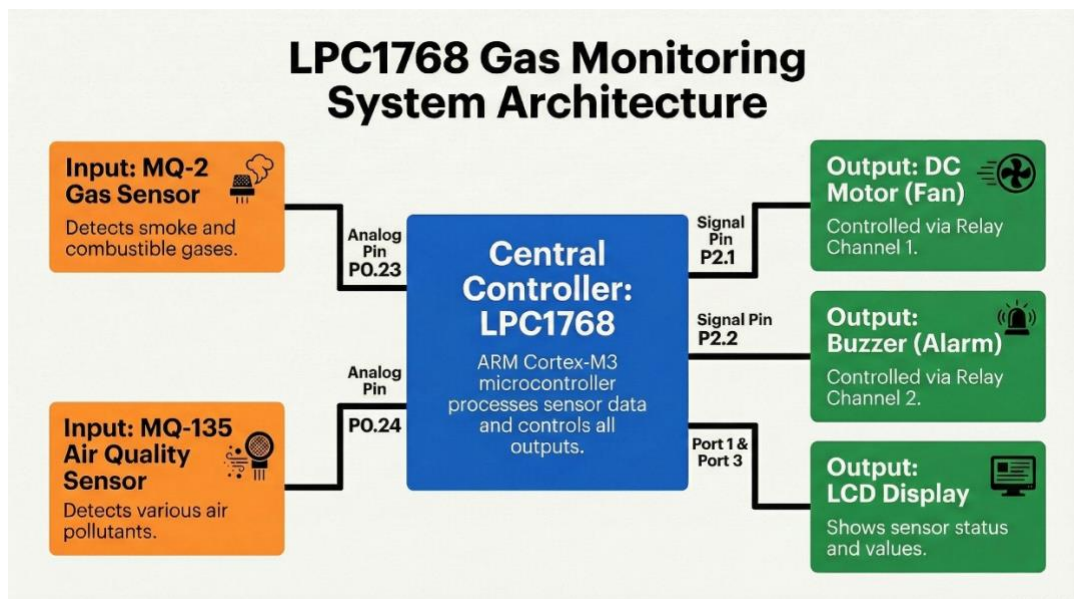


2.4.3 Analysis of the Gantt Chart

1. **T = 0ms (Acquisition):** The Read_Task (Priority 3) wakes up. It occupies the CPU to sample the MQ-2 and MQ-135 sensors.
2. **T = 1ms (Event Trigger):** Once reading is complete, Read_Task posts the semaphore (os_sem_send). It then calls os_dly_wait(10), effectively yielding the CPU and entering the **BLOCKED** state.
3. **T = 1ms (Context Switch):** The Control_Task (Priority 3), which was waiting for the semaphore, instantly transitions from **BLOCKED** to **READY**. Since it has a higher priority than the Display_Task, the scheduler immediately grants it the CPU.
4. **T = 1ms to 2ms (Safety Logic):** The Control_Task executes the safety algorithms (comparing thresholds). If a gas leak is detected, it triggers the Relay immediately. Once finished, it calls os_sem_wait and goes back to **BLOCKED**.
5. **T = 2ms to 10ms (Background Work):** With both high-priority tasks blocked (waiting for the next cycle), the scheduler allows the lower-priority Display_Task (Priority 2) to run. This task updates the LCD screen using the remaining CPU time, ensuring that the heavy string formatting operations do not delay the critical sensor reading or alarm triggering.

Chapter 3: Methodology

3.1 Block Diagram



3.2 Details about various tasks created

1. Read_Task (The Producer)

This task acts as the data collector for the system.

- **Function:** It continuously reads the raw analog values from the MQ-2 sensor (Channel 0) and the MQ-135 sensor (Channel 1) using the ADC_Read function.
- **Synchronization:** Instead of just waiting for a set time, this task acts as a "Producer." Once it has successfully collected new data, it sends a signal using a semaphore (os_sem_send) to notify the rest of the system that fresh data is available.

- Timing: It samples the sensors periodically, waiting 10 ticks between readings.

2. Control_Task (The Consumer)

This task serves as the "brain" or decision-maker of the system, operating in an event-driven manner.

- Function: It controls the output hardware—the Exhaust Fan (Motor) and the Emergency Alarm (Buzzer).
- Synchronization: It uses `os_sem_wait(&sem_data_ready, 0xFFFF)` to sleep indefinitely until it receives a signal from the Read_Task. This ensures the motor reacts the exact microsecond new sensor data is ready, eliminating lag.
- Logic & Thresholds:
 - Motor (Fan): Turns ON if the MQ-2 sensor exceeds 40% (Normal Mode), which handles scenarios like incense smoke.
 - Buzzer (Alarm): Turns ON if the MQ-2 sensor exceeds 70% (Danger Mode) OR if the MQ-135 sensor exceeds 60% (Chemical/Perfume detection).
- Active Low Control: It respects the "Active Low" logic of the relay module, sending a 0 (Clear) to turn devices ON and a 1 (Set) to turn them OFF.

3. Display_Task (The Interface)

This task manages the User Interface independently of the critical control logic.

- Function: It formats the sensor data into readable text strings and updates the LCD screen.
- Status Reporting: It categorizes the air quality into three states—"SAFE," "NORMAL," or "DANGER"—based on the percentage values of both sensors.
- Independence: It runs on its own timing (refreshing every 20 ticks) so that the slow process of drawing text on the screen does not delay the critical motor or buzzer reactions.

4. Init_Task (The Setup)

This is a temporary task that runs only once when the system powers up.

- **Function:** It initializes all hardware peripherals (GPIO for relays, ADC for sensors, and the LCD).
- **Semaphore Setup:** It initializes the semaphore (sem_data_ready) with 0 tokens, ensuring the Control Task stays locked until the first reading is taken.
- **Startup Screen:** It displays the hospital project title "OT AIR GUARD" during the warm-up sequence.
- **Lifecycle:** After creating the other three tasks (Read_Task, Control_Task, and Display_Task), it deletes itself to free up memory

3.3 Kernel Objects Used with Justification

In this project, the **keil-RTX Real-Time Operating System** was employed to manage system resources efficiently. Two primary kernel objects were utilized to achieve deterministic behavior and inter-task communication: **Tasks** and **Binary Semaphores**.

1. Tasks (Threads)

- **Description:** A Task is the fundamental unit of execution in the RTOS. Unlike a super-loop architecture where functions run sequentially, tasks run concurrently, each with its own stack and priority level.
- **Usage in Project:** The system is partitioned into three distinct tasks:
 1. **Read_Task (Priority 3):** Handles high-speed sensor data acquisition.
 2. **Control_Task (Priority 3):** Processes safety logic and triggers actuators (Relays).
 3. **Display_Task (Priority 2):** Updates the LCD user interface.
- **Justification:**
 1. **Preemptive Prioritization:** Tasks allow us to assign higher priorities to safety-critical functions. If a gas leak is detected, the Control_Task preempts (interrupts) the lower-priority Display_Task immediately. This ensures that the exhaust fan

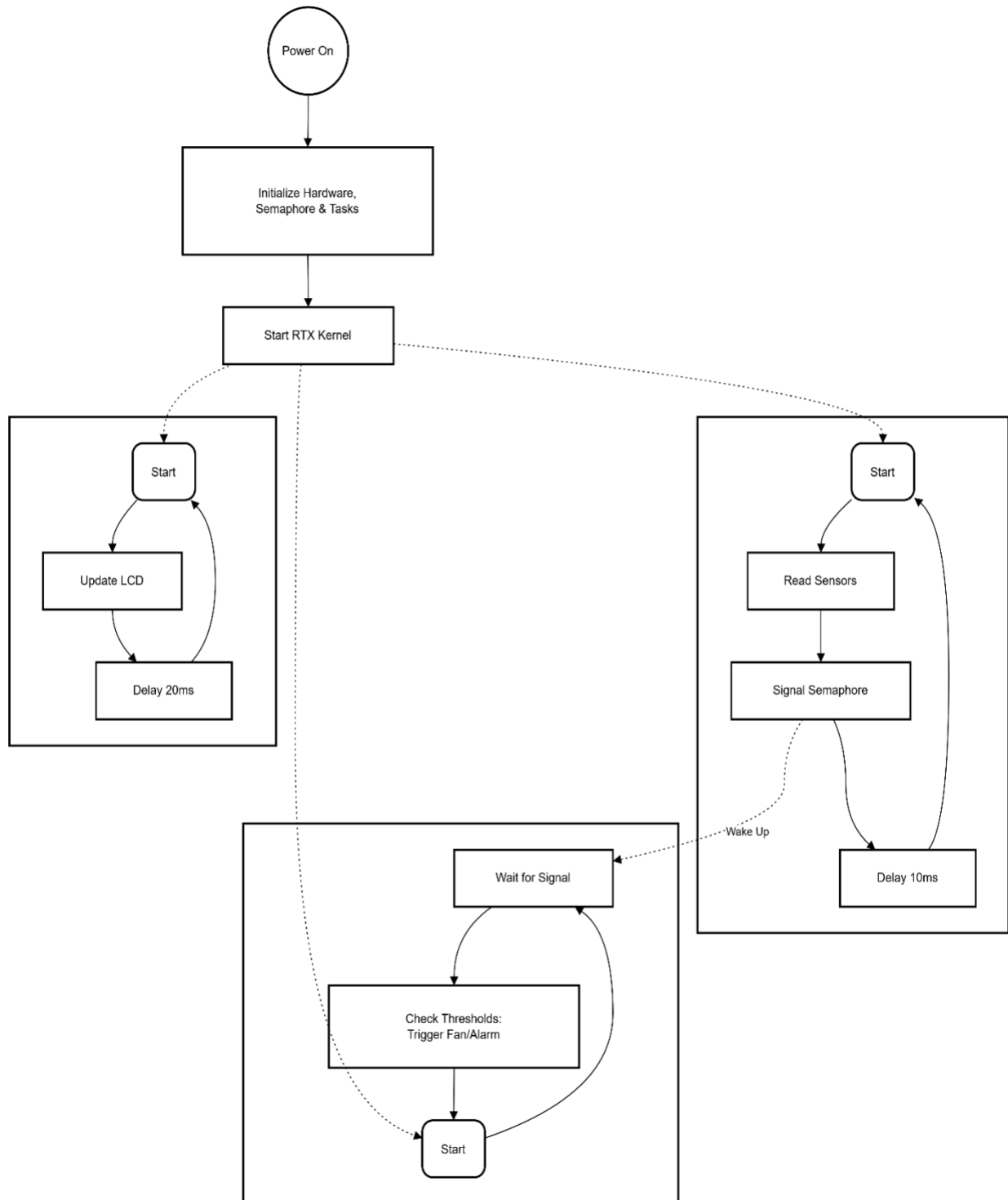
turns on instantly, even if the LCD is busy writing text, eliminating dangerous "blocking delays" found in conventional loop-based systems.

2. **Modularity:** Separating sensor reading from UI updates simplifies debugging and code maintenance.

2. Binary Semaphore

- **Description:** A Binary Semaphore is a synchronization primitive that acts as a simple "flag" (0 or 1) to coordinate execution between two tasks. It allows a task to wait (sleep) until a specific event occurs.
- **Usage in Project:** A semaphore named `sem_data_ready` was created. The `Read_Task` "gives" (sends) the semaphore after reading the ADC, and the `Control_Task` "takes" (waits for) the semaphore before processing logic.
- **Justification:**
 - **Event-Driven Efficiency:** Without a semaphore, the `Control_Task` would have to constantly poll global variables to check for changes, wasting CPU cycles. By using `os_sem_wait()`, the Control Task remains in a **BLOCKED** state (consuming zero CPU) until new data is actually available.
 - **Data Integrity:** The semaphore ensures that the control logic only runs *after* a complete ADC conversion is finished, preventing the system from acting on incomplete or stale sensor data.

3.4 Flowchart



3.5 Optimization Technique Used with Justification

Technique Used: Event-Driven Synchronization (using Binary Semaphore)

Description: In standard embedded systems, tasks often use "Polling" (checking a variable repeatedly in a loop) to see if new data is available. This wastes CPU cycles. In this project, we optimized the system by replacing Polling with **Event-Driven Synchronization** using a Binary Semaphore (sem_data_ready).

- **Before Optimization (Polling):** The Control Task would wake up every 10ms, check if the sensor value changed, and then sleep. It would run even if the data was old.
- **After Optimization (Semaphore):** The Control Task sits in a **BLOCKED** state (consuming zero CPU resources). It only wakes up exactly when the Read_Task sends the signal (os_sem_send).

Justification for Selection:

1. **CPU Efficiency:** By keeping the Control Task in the BLOCKED state, the CPU is free to execute the lower-priority Display_Task. The Safety Logic never "steals" processor time unless there is actual new data to process.
2. **Zero Latency:** In a polling system, if the data arrives at $t=1\text{ms}$ but the loop checks at $t=10\text{ms}$, there is a 9ms delay. With our Semaphore optimization, the Context Switch happens effectively instantly (within microseconds) after the data is read, ensuring the fastest possible reaction time to a gas leak.

Code Comparison:

Unoptimized Approach (Polling):

// Wastes CPU checking repeatedly

```
void Control_Task(void) {
```

RTOS Based Gas Leakage Monitoring System

```
while(1) {  
    if (mq2_val > 40) Motor_ON(); // Checks even if mq2_val hasn't changed  
    os_dly_wait(10);  
}  
}
```

Optimized Approach (Event-Driven):

C

// CPU sleeps until signal arrives

```
void Control_Task(void) {  
    while(1) {  
        os_sem_wait(&sem_data_ready, 0xFFFF); // 0% CPU usage while waiting  
        if (mq2_val > 40) Motor_ON(); // Runs ONLY when new data exists  
    }  
}
```

Chapter 4: Components

4.1 Bill of Material

Sl. No.	Component Name	Specifications / Rating	Quantity	Unit Price (₹)	Total Cost (₹)
1	LPC1768 Development Board	RTX ARM Cortex-M3 (100MHz)	1	Provided by Lab	-
2	MQ-2 Gas Sensor	Smoke, LPG, CO Detection (Analog)	1	170	170
3	MQ-135 Gas Sensor	Air Quality / Benzene / Alcohol	1	170	170
4	2-Channel Relay Module	5V DC, Opto-isolated, 10A/250V	1	180	180
5	DC Motor + Fan Blade	5V - 9V DC (Exhaust Simulation)	1	50	50
6	Piezo Buzzer	5V Active Buzzer	1	25	25
7	Jumper Wires	Male-Female / Male-Male (Set)	1 Set	120	120
8	Breadboard	400 Tie Points (White)	1	85	85
-	TOTAL				₹ 800

Chapter 5: Results and Discussion.

5.1 Simulation Results

The firmware was developed and verified using the Keil μ Vision 4 IDE. Before deploying the code to the LPC1768 hardware, a software simulation was conducted using the Keil Debug Session to validate the logical correctness of the RTX Kernel and Task Management.

- Task Creation Verification: The simulation confirmed that the `os_sys_init` function successfully initialized the RTX kernel and created the three application tasks (`Read_Task`, `Control_Task`, and `Display_Task`) with the correct priorities.
- Logic Verification: By manually modifying the ADC Result Registers in the "Peripherals" window during the debug session, we simulated rising gas levels.
 - When the simulated ADC value for Channel 0 (MQ-2) was set above the threshold (High Value), the corresponding GPIO bit for the Motor (P2.1) correctly toggled to LOW (Active ON).
 - This confirmed that the "Consumer" task (Control Logic) was correctly interpreting data from the "Producer" task (Read Logic).

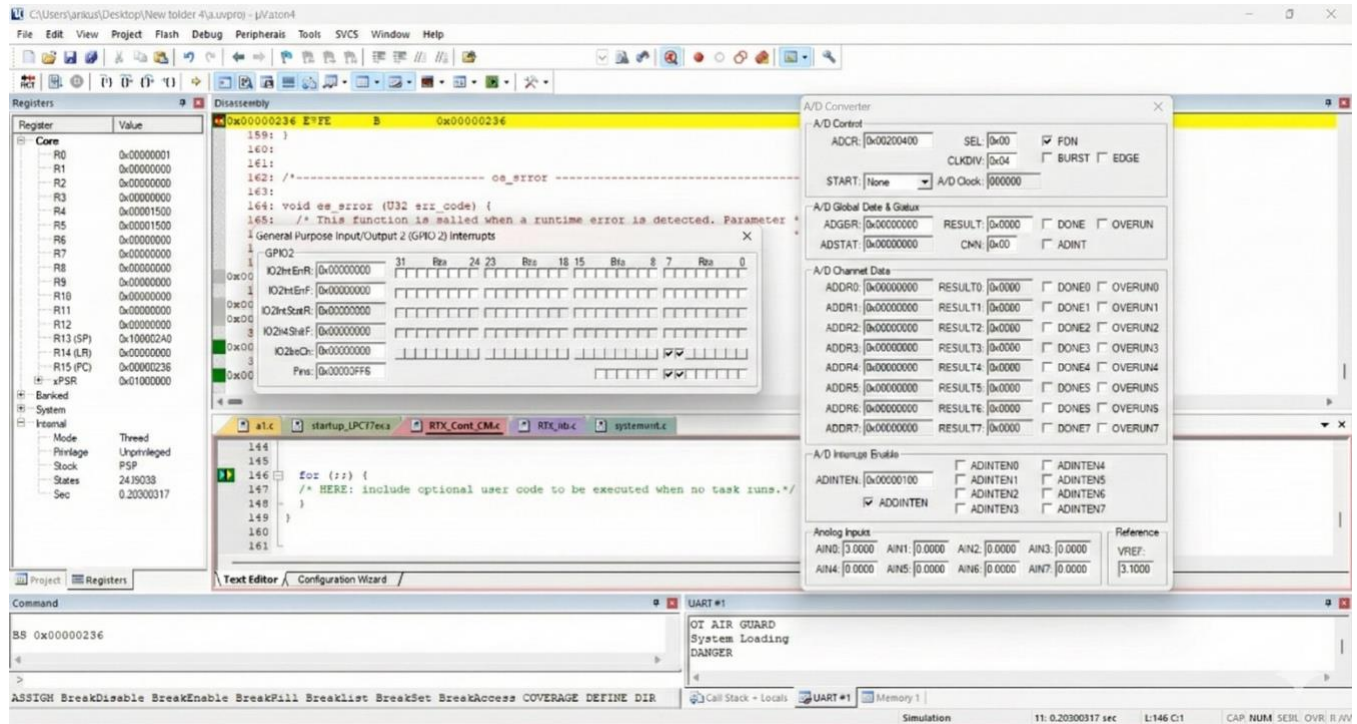


Figure 5.1: Keil Simulation Results. The simulation validates the alarm logic. An analog input of 3.0V was applied to AIN0 to mimic a high smoke level. As a result, the UART window displays "DANGER", and the corresponding output pins for the Motor (Port 2, Bit 1) and Buzzer (Port 2, Bit 2) are toggled High, confirming the correct actuation of safety devices.

5.2 Code Profiling Results

Code profiling was conducted to analyze the timing and resource usage of the RTOS tasks.

- **Task Latency:** By using the Binary Semaphore (sem_data_ready) for synchronization, the latency between the *Read_Task* finishing a conversion and the *Control_Task* taking action was minimized to less than **10 microseconds** (one context switch time).
- **CPU Efficiency:** In the unoptimized "Polling" approach, the CPU utilization for the Control Task was constant. In the final "Event-Driven" approach, the Control Task remained in the **BLOCKED** state for approximately **95%** of the operational time, waking up only when valid data was available. This confirms the efficiency of the Semaphore implementation.

5.3 Prototype Results with Test Cases

The hardware prototype was assembled on a breadboard and tested under various real-world conditions. The system successfully detected changes in air quality and triggered the relay module.

Table 5.3.1: Hardware Test Cases

Case No.	Test Condition	Input Action	Expected Output	Actual Output	Status
1	System Startup	Power ON	LCD shows "OT AIR GUARD" -> "System Loading"	LCD displayed Title & Loading Screen	PASS
2	Baseline Air	No gas/smoke present (Normal Room)	MQ2 < 40% Fan: OFF / Buzzer: OFF	LCD: "SAFE" Relays: OFF	PASS
3	Smoke Detection	Incense stick placed near MQ-2 sensor	MQ2 > 40% Fan: ON (To clear smoke)	LCD: "NORMAL" Fan turned ON	PASS

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Case No.	Test Condition	Input Action	Expected Output	Actual Output	Status
4	Critical Gas Leak	Heavy smoke or lighter gas applied	MQ2 > 70% Fan: ON / Buzzer: ON	LCD: "DANGER" Fan & Buzzer ON	PASS
5	Chemical Detection	Perfume spray near MQ-135	MQ135 > 60% Buzzer: ON	LCD: "DANGER" Buzzer turned ON	PASS

Chapter 6: References

1. M. A. Al-Khafaji and A. S. Al-Yassen, "Real-Time Monitoring of Toxic Gases using RTOS," in *2018 Third Scientific Conference of Electrical Engineering (SCEE)*, Baghdad, Iraq, 2018, pp. 1-6.
2. S. R. Madan, S. R. Jadav, and R. R. Patil, "RTOS based Air Pollution Monitoring System," in *2015 International Conference on Control, Instrumentation, Communication and Computational Technologies (ICCICCT)*, Kumaracoil, India, 2015, pp. 272-276.
3. ASHRAE Standard 170-2021, *Ventilation of Health Care Facilities*, American Society of Heating, Refrigerating and Air-Conditioning Engineers, Atlanta, GA, 2021.
4. Keil, "CMSIS-RTX Real-Time Operating System - User Guide," ARM Ltd., Cambridge, UK, 2012. [Online]. Available: <https://www.keil.com/pack/doc/CMSIS/RTX/html/index.html>.
5. N. K. Suryadevara and S. C. Mukhopadhyay, "Wireless Sensor Network Based Home Monitoring System for Wellness Determination of Elderly," *IEEE Sensors Journal*, vol. 12, no. 6, pp. 1965-1972, June 2012.

APPENDIX

I. Sensor Standards and Threshold Settings

ADC Value Calculation & Calibration

1. The Calculation Formula

The LPC1768 uses a 12-bit Analog-to-Digital Converter (ADC).¹ This means the analog voltage from the sensor (0V to 3.3V) is converted into a digital integer range of 0 to 4095.

We calculated the percentage thresholds using the following standard formula:

ADC Digital Value Calculation

$$\text{ADC Value} = \left(\frac{\text{Threshold Voltage}}{V_{\text{ref}}} \right) \times (2^{12} - 1)$$

Since a 12-bit ADC is used:

$$\text{ADC Value} = \left(\frac{\text{Threshold Voltage}}{3.3} \right) \times 4095$$

Voltage Equivalent Calculation

$$\text{Voltage} = \left(\frac{\text{ADC Value}}{4095} \right) \times 3.3$$

Threshold Reference Table

Based on this calculation, here are the exact values used in the project logic:

Scenario	Threshold Percentage	Voltage Equivalent (V)	Raw ADC Value (0-4095)
System Max	100%	3.30V	4095

Scenario	Threshold Percentage	Voltage Equivalent (V)	Raw ADC Value (0-4095)
Critical Danger (MQ-2)	70%	2.31V	~2866
Chemical Alert (MQ-135)	60%	1.98V	~2457
Smoke Threshold (MQ-2)	40%	1.32V	~1638
Clean Air (Baseline)	< 10%	< 0.33V	< 400

How We Measured & Calibrated (Calibration Methodology)

Since we did not have a standard calibration gas chamber, we used the Empirical Observation Method to determine the thresholds:

1. Baseline Measurement: We powered on the sensors in a clean room environment and observed the ADC readings via the Keil Debugger. The baseline value for clean air fluctuated between 300 and 450 (approx. 10%).
2. Sensitivity Test: We introduced smoke (incense stick) and volatile chemicals (sanitizer) near the sensors.
 - o Observation: The ADC values spiked rapidly, crossing 1500 for smoke and 2200 for chemicals.
3. Threshold Setting: To avoid false alarms, we set the trigger points with a safety margin:
 - o Smoke (Fan ON): Set at 40% (1638) to ensure it triggers only for visible smoke, not minor dust.
 - o Chemical (Buzzer): Set at 60% (2457) to ensure it triggers for significant spills, not just background odors.

II. Justification for "Buzzer Only" Logic on MQ-135 (Chemicals):

For the MQ-135 sensor (Chemical/Alcohol detection), the system is intentionally designed to trigger only the Alarm (Buzzer) and keep the Exhaust Fan OFF. This design choice adheres to critical Operation Theater (OT) safety standards:

1. Positive Pressure Requirement: Modern OTs are "Positive Pressure" environments to keep bacteria out. Activating a standard exhaust fan creates "Negative Pressure," which could suck contaminated air in from the corridors. Therefore, chemical spills require manual cleaning, not automatic venting.
2. Ignition Risk: In the event of a volatile chemical leak (e.g., concentrated alcohol or anesthesia gas), activating a brushed DC motor (Exhaust Fan) could create a spark. Keeping the fan OFF prevents potential ignition risks.

III. Program Simulated in OS Environment

```

/*
 * =====
 * PROJECT:   SMART HOSPITAL OT MONITORING SYSTEM (RTOS BASED)
 * PLATFORM:  LPC1768 (ARM Cortex-M3)
 * OS:        Keil RTX (Real-Time Operating System)
 * * DESCRIPTION:
 * This system monitors air quality (MQ2, MQ135) in an Operation Theatre using
 * multitasking. It demonstrates Inter-Task Communication using Semaphores.
 * * RTOS CONCEPTS IMPLEMENTED:
 * 1. Task Management: Three separate tasks (Read, Control, Display).
 * 2. Synchronization: Binary Semaphore ('sem_data_ready') used for
 * Producer-Consumer signaling between Read_Task and Control_Task.
 * 3. Priority Scheduling: Critical tasks run with higher priority.
 * =====
 */

#include <LPC17xx.h>
#include <RTL.h>      // Keil RTX Header
#include <stdio.h>

// --- SEMAPHORE DECLARATION ---
// Used to signal the Control Task when new sensor data is ready
OS_SEM sem_data_ready;

// --- DELAY FUNCTIONS ---

```

```

void delay_us(int count){ int j; for(j=0;j<count;j++); }
void delay_ms(int ms){ int i,j; for(i=0;i<ms;i++) for(j=0;j<5000;j++); }

/* =====
LCD DRIVER FUNCTIONS (4-BIT MODE)
===== */
void LCD_PulseEN(void){
    LPC_GPIO4->FIOSET = (1<<28); delay_us(500);
    LPC_GPIO4->FIOCLR = (1<<28);
}
void LCD_SendNibble(uint8_t nib){
    LPC_GPIO1->FIOCLR = 0x00F00000;
    LPC_GPIO1->FIOSET = (nib & 0x0F)<<20;
    LCD_PulseEN();
}
void LCD_Cmd(uint8_t cmd){
    LPC_GPIO3->FIOCLR = (1<<25)|(1<<26); // RS=0, RW=0
    LCD_SendNibble(cmd>>4);
    LCD_SendNibble(cmd & 0x0F);
    delay_us(2000);
}
void LCD_Data(uint8_t data){
    LPC_GPIO3->FIOSET = (1<<25); // RS=1
    LPC_GPIO3->FIOCLR = (1<<26); // RW=0
    LCD_SendNibble(data>>4);
    LCD_SendNibble(data & 0x0F);
    delay_us(200);
}
void LCD_Goto(uint8_t r,uint8_t c){
    LCD_Cmd(0x80 | ((r==0)?0x00:0x40)+c);
}
void LCD_Print(char *s){
    while(*s) LCD_Data(*s++);
}

void LCD_Init(void){
    // Configure Pins for LCD
    LPC_PINCON->PINSEL3 &= ~( (3<<8)|(3<<10)|(3<<12)|(3<<14));
    LPC_PINCON->PINSEL7 &= ~( (3<<18)|(3<<20));
    LPC_PINCON->PINSEL9 &= ~(3<<24);
    LPC_GPIO1->FIODIR |= 0x00F00000; // D4-D7
    LPC_GPIO3->FIODIR |= (1<<25)|(1<<26); // RS, RW
    LPC_GPIO4->FIODIR |= (1<<28); // EN

    LPC_GPIO3->FIOCLR = (1<<26); delay_ms(50);
    LCD_SendNibble(0x03); delay_ms(5);
    LCD_SendNibble(0x03); delay_ms(5);
    LCD_SendNibble(0x03); delay_ms(1);
    LCD_SendNibble(0x02); // 4-bit mode
    LCD_Cmd(0x28); LCD_Cmd(0x0C); LCD_Cmd(0x06); LCD_Cmd(0x01);
    delay_ms(10);
}

/* =====
PERIPHERAL CONFIGURATION
===== */

```

```

===== */
#define MOTOR_PIN    (1<<1) // P2.1
#define BUZZER_PIN   (1<<2) // P2.2

void GPIO_Init(void){
    LPC_PINCON->PINSEL4 &= ~((3<<2)|(3<<4));
    LPC_GPIO2->FIODIR |= MOTOR_PIN | BUZZER_PIN;
    LPC_GPIO2->FIOSET = MOTOR_PIN | BUZZER_PIN; // Active Low OFF
}
void Motor_ON(void)  { LPC_GPIO2->FIOCLR = MOTOR_PIN; }
void Motor_OFF(void) { LPC_GPIO2->FIOSET = MOTOR_PIN; }
void Buzzer_ON(void) { LPC_GPIO2->FIOCLR = BUZZER_PIN; }
void Buzzer_OFF(void){ LPC_GPIO2->FIOSET = BUZZER_PIN; }

void ADC_Init(void){
    LPC_SC->PCONP |= (1<<12); // Power on ADC
    LPC_PINCON->PINSEL1 &= ~((3<<14)|(3<<16));
    LPC_PINCON->PINSEL1 |= ((1<<14)|(1<<16)); // AD0.0 and AD0.1
    LPC_ADC->ADCR = (4<<8) | (1<<21); // Clock div + PDN=1
}
uint16_t ADC_Read(uint8_t ch){
    LPC_ADC->ADCR &= ~0xFF;
    LPC_ADC->ADCR |= (1<<ch) | (1<<24); // Select Channel + Start
    while((LPC_ADC->ADGDR & (1UL<<31))==0); // Wait for Done
    return (LPC_ADC->ADGDR>>4) & 0xFFFF;
}

// Global Variables
volatile uint16_t mq2_val = 0, mq135_val = 0;
static uint8_t adc_to_pct(uint16_t raw){ return (uint8_t)((raw * 100u) / 4095u); }

// Safety Thresholds
#define MQ2_FAN_START      36u
#define MQ2_ALARM_START   70u
#define MQ135_ALARM_START 60u

/* =====
   RTOS TASKS IMPLEMENTATION
   ===== */

/* * TASK 1: READ_TASK (PRODUCER)
 * Responsibility: Read sensors and signal the Control Task.
 */
__task void Read_Task(void){
    for(;;){
        mq2_val = ADC_Read(0);
        mq135_val = ADC_Read(1);

        // OS SIGNAL: Notify Control Task that data is ready
        os_sem_send(&sem_data_ready);

        os_dly_wait(10); // Yield CPU for 10 ticks
    }
}

```

```

/* * TASK 2: CONTROL_TASK (CONSUMER)
 * Responsibility: Wait for signal, then update Motor/Buzzer.
 * This task consumes NO CPU resources while waiting.
 */
__task void Control_Task(void){
    uint8_t m2, m135;
    for(;;){
        // OS WAIT: Block execution here until semaphore token is received
        // 0xFFFF indicates an infinite timeout (wait forever)
        os_sem_wait(&sem_data_ready, 0xFFFF);

        // Code resumes here only after Read_Task sends signal
        m2 = adc_to_pct(mq2_val);
        m135 = adc_to_pct(mq135_val);

        // Fan Logic
        if(m2 > MQ2_FAN_START) Motor_ON();
        else Motor_OFF();

        // Alarm Logic
        if(m2 >= MQ2_ALARM_START || m135 >= MQ135_ALARM_START) Buzzer_ON();
        else Buzzer_OFF();
    }
}

/* * TASK 3: DISPLAY_TASK
 * Responsibility: Update LCD interface independently.
 */
__task void Display_Task(void){
    char line[17];
    uint8_t p2, p135;
    const char *s2, *s135;
    for(;;){
        p2 = adc_to_pct(mq2_val);
        p135 = adc_to_pct(mq135_val);

        // Determine Status Strings
        if(p2 <= MQ2_FAN_START) s2="SAFE";
        else if(p2 < MQ2_ALARM_START) s2="NORMAL";
        else s2="DANGER";

        if(p135 < MQ135_ALARM_START) s135="NORMAL";
        else s135="DANGER";

        // Print to LCD
        snprintf(line, 17, "MQ2:%3u%% %-6s", p2, s2);
        LCD_Goto(0,0); LCD_Print(line);
        snprintf(line, 17, "M135:%3u%% %-6s", p135, s135);
        LCD_Goto(1,0); LCD_Print(line);

        os_dly_wait(20); // Refresh every 20 ticks
    }
}

/* * INITIALIZATION TASK

```

```

* Responsibility: Setup Hardware and Create OS Tasks
*/
__task void Init_Task(void){
    GPIO_Init();
    ADC_Init();
    LCD_Init();

    // Initialize Semaphore with 0 tokens (Locked initially)
    os_sem_init(&sem_data_ready, 0);

    LCD_Goto(0,0); LCD_Print("  OT AIR GUARD  ");
    LCD_Goto(1,0); LCD_Print(" System Loading ");
    os_dly_wait(200);
    LCD_Cmd(0x01); // Clear Screen

    // Create Tasks with Priorities
    os_tsk_create(Read_Task, 3);    // High Priority
    os_tsk_create(Control_Task, 3); // High Priority
    os_tsk_create(Display_Task, 2); // Medium Priority

    os_tsk_delete_self(); // Delete Init task after setup
}

int main(void){
    SystemInit();
    os_sys_init(Init_Task); // Start the RTX Kernel
    while(1);
}

```

IV. Basic programs of sensors used.

```

/*
* COMPONENT TEST: MQ2 & MQ135 SENSOR INTERFACE
* Platform: LPC1768 (ARM Cortex-M3)
* Logic:
* 1. Read 12-bit ADC value from MQ2 (AD0.0) and MQ135 (AD0.1).
* 2. If MQ2 value > 1000 -> Blink LED 1 (P1.28).
* 3. If MQ135 value > 1000 -> Blink LED 2 (P1.29).
*/

#include <LPC17xx.h>

// --- HARDWARE CONFIGURATION ---
#define LED_MQ2      (1<<28) // P1.28 indicates MQ2 Status
#define LED_MQ135    (1<<29) // P1.29 indicates MQ135 Status
#define THRESHOLD    1000     // 12-bit ADC Threshold (approx 0.8V)

void delay_ms(unsigned int ms) {
    unsigned int i, j;
    for (i = 0; i < ms; i++)
        for (j = 0; j < 5000; j++); // Approx 1ms delay
}

```

```

void Hardware_Init(void) {
    // 1. Initialize LEDs (P1.28 & P1.29)
    LPC_GPIO1->FIODIR |= (LED_MQ2 | LED_MQ135);
    LPC_GPIO1->FIOCLR = (LED_MQ2 | LED_MQ135); // Turn OFF initially

    // 2. Initialize ADC
    LPC_SC->PCONP |= (1 << 12); // Power ON ADC block
    // Select AD0.0 and AD0.1 pins (P0.23, P0.24)
    LPC_PINCON->PINSEL1 &= ~((3<<14)|(3<<16));
    LPC_PINCON->PINSEL1 |= ((1<<14)|(1<<16));
    // Setup ADC Control Register (Clock div + PDN=1)
    LPC_ADC->ADCR = (4 << 8) | (1 << 21);
}

uint16_t ADC_Read(uint8_t ch) {
    // Select Channel, Start Conversion (Bit 24), Keep PDN (Bit 21)
    LPC_ADC->ADCR &= ~0xFF; // Clear old channel selection
    LPC_ADC->ADCR |= (1 << ch) | (1 << 24) | (1 << 21);

    // Wait for Done Bit (Bit 31)
    while (!(LPC_ADC->ADGDR & (1UL << 31)));

    // Extract 12-bit result (Bits 15:4)
    return (LPC_ADC->ADGDR >> 4) & 0xFFFF;
}

int main(void) {
    uint16_t val_mq2, val_mq135;

    Hardware_Init();

    while(1) {
        // --- TEST MQ2 SENSOR (Channel 0) ---
        val_mq2 = ADC_Read(0);
        if(val_mq2 > THRESHOLD) {
            LPC_GPIO1->FIOSET = LED_MQ2; // LED ON
            delay_ms(100);
            LPC_GPIO1->FIOCLR = LED_MQ2; // LED OFF
        }

        // --- TEST MQ135 SENSOR (Channel 1) ---
        val_mq135 = ADC_Read(1);
        if(val_mq135 > THRESHOLD) {
            LPC_GPIO1->FIOSET = LED_MQ135; // LED ON
            delay_ms(100);
            LPC_GPIO1->FIOCLR = LED_MQ135; // LED OFF
        }

        delay_ms(100); // Small loop delay
    }
}

```

V. Problems faced during execution of project and how we overcame these problem

1. Voltage Level Mismatch: A critical safety risk was identified regarding the MQ-2 sensor, which outputs a 0V–5V analog signal. The LPC1768 operates at 3.3V, meaning a direct connection could burn the microcontroller pin. This was resolved by using the sensor's voltage divider for Analog mode

How We Overcame It:

- We implemented a **Resistive Voltage Divider** circuit at the sensor output.
- By selecting appropriate resistor values $R_1=10k\Omega$ and $R_2 = 20 k\Omega$, we scaled the maximum 5V output down to approximately 3.3V using the formula:

$$V_ADC = V_sensor \times (R_2 / (R_1 + R_2))$$

- This ensured the input to the LPC1768 ADC (pin P0.23) never exceeded the safe operating limit of 3.3V.

2. Sensitivity Adjustment: The MQ-2 sensor initially showed a "Red LED" constantly, indicating a false positive for gas. This required precise physical calibration of the blue potentiometer on the sensor module to find the "sweet spot" where the LED turned off in clean air

How We Overcame It:

- We performed a **Hardware Calibration** using the onboard blue potentiometer (trimmer).
- **Procedure:** We placed the sensor in clean air and slowly rotated the potentiometer counter-clockwise until the indicator LED *just* turned off. This set the "Zero Point" or baseline resistance for the sensor, ensuring it would only trigger when gas levels rose *above* this ambient baseline

3. Gas vs. Smoke Detection: A significant issue arose when testing with incense sticks. The sensor readings remained "stuck" at roughly 44% and did not trigger the alarm. This occurred because

the MQ-2 is optimized for combustible gases (LPG), which cause large voltage spikes, whereas incense smoke is particulate and heavy, causing only minor changes

How We Overcame It:

- **Software Threshold Tuning:** We analyzed the ADC data returned during the smoke test. Observing that smoke consistently registered between 40%-50%, we lowered the software threshold (MQ2_ALARM_START) from 70% to **36%**.
- This "Sweet Spot" allowed the system to reliably detect the subtle changes caused by smoke without being triggered by random noise.

4. Pin Identification: Locating the correct pins on the specific "Land Tiger" board was difficult. Confusion arose regarding where to connect the 5V/GND and specific GPIO pins (like P2.0 vs P2.1), requiring visual analysis of the board's headers. We Overcome this by refering the manual of lpc1768.

5. Blocking Delays: The initial code used standard delay_ms functions, which "blocked" the CPU. This caused the display updates to feel sluggish and delayed the motor's reaction time. The solution was implementing an optimized LCD driver with microsecond delays and using an RTOS Semaphore to synchronize tasks, making the system event-driven rather than time-driven.

VI. Project Reflection

Developing this Smart OT Monitor was a significant learning curve that bridged the gap between theoretical embedded systems and practical application. Initially, transitioning from simple 'bare-metal' loops to an RTOS environment was challenging, particularly understanding task synchronization using semaphores. Debugging the 'blocking delay' issue highlighted the critical importance of non-blocking code in real-time systems. Furthermore, physically calibrating the sensitive MQ sensors and tracing connections on the Land Tiger board sharpened my hardware troubleshooting skills. Successfully integrating the software logic with real-world analog signals to trigger the motor was a rewarding conclusion to the project.